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RESEARCH ARTICLE

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Robust point and variance estimation for meta-analyses with selective reporting and dependent effect sizes

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Abstract

1. Meta-analysis produces a quantitative synthesis of evidence-based knowledge, shaping not only research trends but also policies and practices in biology. However, two statistical issues, selective reporting and statistical dependence, can severely distort meta-analytic parameter estimation and inference.
2. Here, we re-analyse 448 meta-analyses to demonstrate a new two-step procedure to deal with two common challenges in biological meta-analyses that often occur simultaneously: publication bias and non-independence. First, we employ bias-robust weighting schemes under the generalized least square estimator to obtain average effect sizes that are more robust to selective reporting. We then use cluster-robust variance estimation to account for statistical dependence, reducing bias in estimating standard errors and ensuring valid statistical inference.
3. The first step of our approach demonstrates comparable performance in estimating average effect sizes to the existing publication-bias adjustment methods in the presence of selective reporting. This equivalence holds across two publication bias selection processes. The second step achieves estimates of standard errors consistent with the multilevel meta-analytic model, a benchmark method with adequate control of Type I error rates for multiple, statistically dependent effect sizes. Re-analyses of 448 meta-analyses show that ignoring these two issues tends to overestimate effect sizes by an average of 110% and underestimate standard errors by 120%.
4. To facilitate implementation, we have developed a website including a step-by-step tutorial. Complementing current meta-analytic workflows with the proposed method as a sensitivity analysis can facilitate a transition to a more robust approach in quantitative evidence synthesis.

KEYWORDS
effect size, evidence synthesis, meta research, meta-analysis, mixed-effects model, publication bias, selective reporting

1 | INTRODUCTION

Quantitative synthesis of research findings has become increasingly important for guiding scientific research and informing evidence-based decision-making (Gurevitch et al., 2018). Meta-analytic modelling is the most commonly used quantitative evidence synthesis method and has been widely applied in ecology and evolution (Nakagawa & Santos, 2012), and related disciplines (Nakagawa, Yang, et al., 2023; Yang et al., 2022). There are numerous statistical models available, but two basic ones are the fixed-effect (FE) and random-effects (RE) models (Borenstein et al., 2010). The FE model assumes that true effect sizes are homogeneous across studies (Borenstein et al., 2021). In contrast, the RE model assumes that true effect sizes are heterogeneous across studies (Viechtbauer, 2007). Despite their popularity, both FE and RE models have limitations in dealing with meta-analytic datasets with complex hierarchical structures, which may result in unreliable parameter estimation (e.g. the point estimate of the overall effect) and statistical inferences, including the null-hypothesis tests and confidence interval construction (Yang et al., 2022).

One major challenge to overcome is statistical dependence among effect size estimates (Cheung, 2019a; Nakagawa & Santos, 2012), which arises mainly due to the presence of multiple, statistically dependent effect size estimates from the same study (Figure 1). For example, multiple estimates per study are a feature of 100% of meta-analyses in environmental sciences (Nakagawa, Yang, et al., 2023) and ecology and evolution (this study), while it is a feature of 89% of meta-analyses in animal science (Yang et al., 2022). Dependency structures can be broadly classified into clustered and correlated designs (with some cases having a mixture of both) (Cheung, 2019a; Yang et al., 2022). Clustered structures arise when effect sizes are nested within a higher-level variable (e.g. the same study and species), while correlated structures arise when the sampling errors are correlated (e.g. longitudinal studies). Failure to account for statistical dependence can lead to underestimated standard errors and high Type I error (false positive) rates (Van den Noortgate et al., 2013). Fortunately, advanced statistical frameworks such as linear mixed-effects models (Kalaian & Raudenbush, 1996; Sheu & Suzuki, 2001; Stram, 1996) and structural equation models (Cheung, 2013, 2014, 2019b; Tu & Wu, 2017) allow one to model dependent effect sizes. For example, the multilevel model, incorporating clusters as random effects, can handle the various sources of dependency (Nakagawa & Santos, 2012; Van den Noortgate et al., 2013), whereas the multivariate model, incorporating correlated random effects and errors, can deal with correlated dependency (Gasparrini et al., 2012; Jackson et al., 2011; Yang et al., 2022). In many disciplines, the multilevel meta-analysis (MLMA) model has become a standard (benchmark) method for dealing with dependent effect sizes due to its flexible random-effects structure (Yang et al., 2022). Meanwhile, a new method called cluster-robust variance estimation (CRVE) is receiving attention (Hedges et al., 2010; Pustejovsky & Tipton, 2018), because it can handle statistical dependence without having to know the exact nature of both clustered and correlated dependency structures (Hedges et al., 2010; Pustejovsky & Tipton, 2018).

While statistical dependence is common, meta-analytic datasets also need to contend with selective reporting which can bias parameter estimation. A well-known example of selective reporting is publication bias, which occurs when there is a tendency to publish only statistically significant findings (Figure 1) (Rosenthal, 1979; Thornton & Lee, 2000). Selective reporting can upwardly bias the point estimate of the population mean effect (Ioannidis et al., 2014). Unfortunately, selective reporting is pervasive not only in ecology and evolution (Kimmel et al., 2023; Yang et al., 2023) but also in medicine, environmental sciences, psychology and economics (Bartoš et al., 2024). Given the high heterogeneity within meta-analyses (Senior et al., 2016), RE models (and their MLMA extensions) are vulnerable to selective reporting because the typical inverse-variance weighting scheme can give equal weight to studies. This process results in less powerful/precise studies contributing more strongly to mean effect estimates, exaggerating the bias driven by selective reporting (Easterbrook et al., 1991; Fanelli et al., 2017). The FE model, despite its tendency to inflate Type I error rates, has the advantage of being less susceptible to publication bias compared to the RE/MLMA models (Bramley et al., 2021; Henmi & Copas, 2010). In addition to the FE model, three other models that are often overlooked, but show resistance to selective reporting, are the unrestricted weighted least square (UWLS) model (Stanley & Doucouliagos, 2017), inverse variance heterogeneity (IVhet) model (Doi et al., 2015) and the Henmi-Copas model (Henmi & Copas, 2010). Furthermore, there are *post-hoc* methods available that specifically aim to adjust for publication bias (Nakagawa et al., 2022). Among these, the precision effect test—precision effect estimate with standard error (PET-PEESE) method stands out as an adjustment method readily usable for correcting for publication bias for statistically dependent effect sizes (Kvarven et al., 2020; Stanley & Doucouliagos, 2014).

Overall, sophisticated methods, such as multilevel and multivariate models, can provide better statistical inferences (e.g. nominal Type I error rates) for statistically dependent effect sizes with realistic data structures. However, such models may overestimate the mean population effect size when selective reporting is present because they weigh studies roughly equally in the presence of high heterogeneity. On the other hand, simple methods that are less vulnerable to selective reporting, such as FE, UWLS, IVhet and Henmi-Copas models (Stanley et al., 2022; Stanley & Doucouliagos, 2014; Stanley & Doucouliagos, 2017), underestimate the standard error,

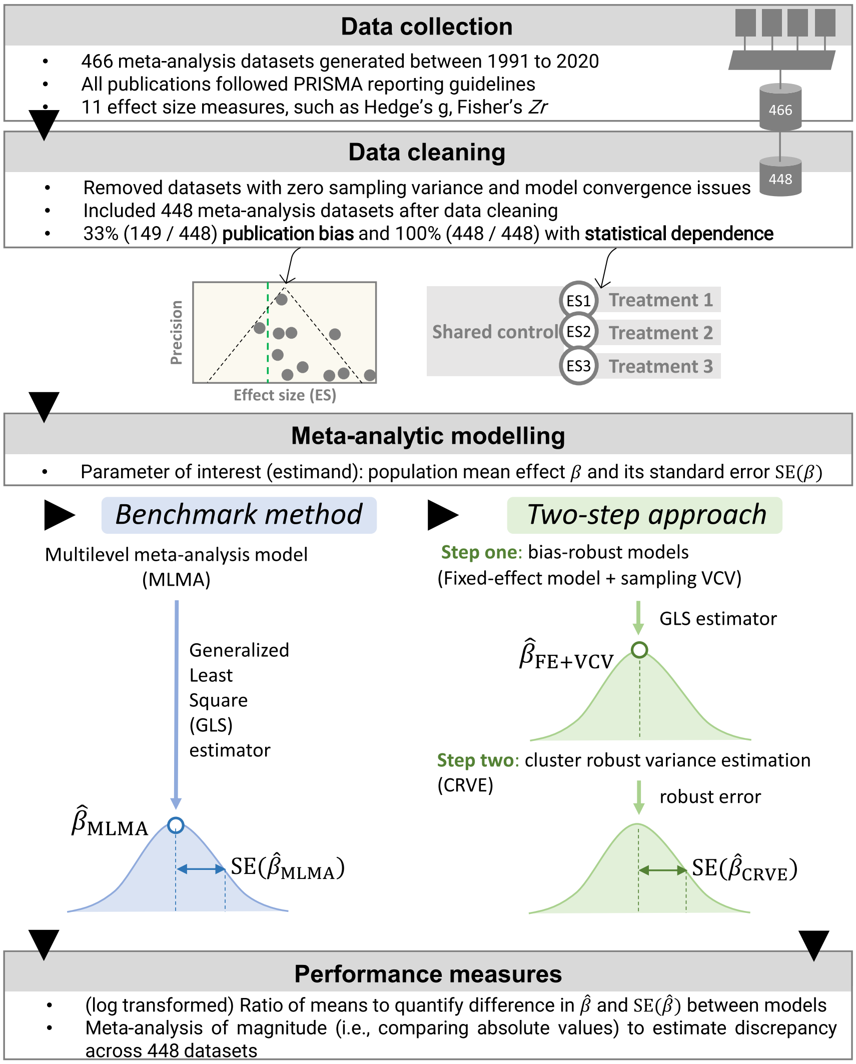

FIGURE 1 Workflow for data compilation, statistical modelling processes and performance criteria. β, $\overline{\beta}$ and $\mathrm{SE}(\beta)$ represent the overall mean effect at the population level, the estimate of the overall effect and the respective standard error.

distorting the statistical inferences (e.g. inflated Type I error rates and reduced p-values). This is because these simple models completely ignore the dependency structure (or more precisely, having a mis-specified variance–covariance structure). Unfortunately, few methods are currently available that can address statistical dependence while also mitigating the impact of publication bias.

Here, we propose an easy-to-implement two-step approach to robustly estimate the mean population effect size and standard error (Figure 1) when selective reporting and statistical dependence occur simultaneously. We have three specific aims. First, we develop the proposed two-step approach in detail. Second, we empirically test the ability of our method to estimate the point estimate relative to PET-PEESE (a publication-bias adjustment method that can account for statistical dependence) using a large dataset consisting of 448 ecological and evolutionary meta-analyses. Third, using the same dataset, we systematically assess the performance of the proposed approach by comparing it to the benchmark method (i.e. MLMA model). Furthermore, we develop and provide R code that can be used to visualize the impacts of selective reporting, providing a new graphical solution for visualizing the impact selective reporting has on parameter estimation and inference, which can be used as a sensitivity analysis.

2 | MATERIALS AND METHODS

To enhance the accessibility of our proposed two-step approach, we minimized the use of formulas as much as possible. For those seeking a mathematical understanding of the principles underlying our method, we refer them to Appendix A.

2.1 | Generalized least squares estimation approach

Before elaborating on our proposed framework, we briefly revisit the generalized least square approach in the context of meta-analytic modelling. Two of the main parameters of interest in meta-analytic modelling are the mean population effect size β (the so-called overall effect) and its standard error $\mathrm{SE}(\beta)$ (square root of variance $\mathrm{Var}(\beta)$).

The overall effect and its standard error can be estimated via two statistical principles (for details, see Appendix A). First, under the normality assumption of the error terms (i.e. random effects and sampling error), iterative generalized least squares (GLS) or equivalently restricted maximum likelihood estimator (REML) yields an unbiased estimator ($\widehat{\beta}$) of the overall effect β (Van Houwelingen et al., 2002):

$$\widehat{\beta} = \left(\sum_{j=1}^{J} \mathbf{1}_j' \mathbf{W}_j \mathbf{1}_j \right)^{-1} \sum_{j=1}^{J} \mathbf{1}_j' \mathbf{W}_j \mathbf{y}_j, \tag{1}$$

where $\mathbf{y}_j = \left(y_{1j}, \ldots, y_{ij}, \ldots, y_{n_j j} \right)'$ denotes an $n_j \times 1$ vector for effect size estimates in j-th study ($i = 1, \ldots, n_j$ and $j = 1, \ldots, J$), $\mathbf{W}_j$ is an $n_j \times n_j$ matrix of weights for the j-th study, and $\mathbf{1}_j$ is the $n_j \times 1$ design matrix consisting of a column of ones allowing estimation of the model intercept.

Second, according to the generalized Gauss-Markov theorem, using the default inverse variance weighting scheme leads to an efficient estimator of the variance of the overall effect $\mathrm{Var}\left(\widehat{\beta}\right)$, meaning a minimal (yet accurate) $\mathrm{Var}\left(\widehat{\beta}\right)$:

$$\begin{aligned} \mathrm{Var}\left(\widehat{\beta}\right) &= \left(\sum_{j=1}^{J} \mathbf{1}_j' \mathbf{W}_j \mathbf{1}_j \right)^{-1} \sum_{j=1}^{J} \mathbf{1}_j' \mathbf{W}_j \boldsymbol{\Sigma}_j \mathbf{W}_j \mathbf{1}_j \left(\sum_{j=1}^{J} \mathbf{1}_j' \mathbf{W}_j \mathbf{1}_j \right)^{-1} \\ &= \left(\sum_{j=1}^{J} \mathbf{1}_j' \mathbf{W}_j \mathbf{1}_j \right)^{-1} \sum_{j=1}^{J} \mathbf{1}_j' \mathbf{W}_j \boldsymbol{\Sigma}_j \boldsymbol{\Sigma}_j^{-1} \mathbf{1}_j \left(\sum_{j=1}^{J} \mathbf{1}_j' \mathbf{W}_j \mathbf{1}_j \right)^{-1} = \left(\sum_{j=1}^{J} \mathbf{1}_j' \boldsymbol{\Sigma}_j^{-1} \mathbf{1}_j \right)^{-1}, \end{aligned} \tag{2}$$

where $\boldsymbol{\Sigma}_j$ represents an $n_j \times n_j$ variance–covariance matrix carrying information about the true statistical dependence structure (where the entries in it capture the covariance between pairs of effect sizes). The inverse of $\boldsymbol{\Sigma}_j$, $\boldsymbol{\Sigma}_j^{-1}$, is the so-called inverse variance weighting scheme. With knowledge of the true statistical dependence structure, we can obtain an unbiased variance $\mathrm{Var}\left(\widehat{\beta}\right)$ and standard error $\mathrm{SE}\left(\widehat{\beta}\right) = \sqrt{\mathrm{Var}\left(\widehat{\beta}\right)}$ for the following-up statistical inferences, including the computation of confidence intervals and null-hypothesis tests (e.g. test statistics, p-value).

While these statistical principles theoretically guarantee an unbiased estimate of the overall effect $\widehat{\beta}$ and standard error $\mathrm{SE}\left(\widehat{\beta}\right)$, they may become invalid in the presence of non-representative meta-analytic datasets, such as those affected by selective reporting and unknown statistical dependency structure. Selective reporting can bias the estimate of β, and statistical dependence can distort the estimate, $\mathrm{SE}\left(\widehat{\beta}\right)$, leading to incorrect statistical inferences (e.g. inflated Type I error rates). In the subsequent sections, we will elaborate on our proposed two-step approach designed to simultaneously address the impact of selective reporting and statistical dependence.

2.2 | Two-step robust point and variance estimation

Our proposed two-step approach employs a bias-robust model within the cluster-robust variance estimation (CRVE) framework (Figure 1). In the first step, we incorporate a bias-robust weighting scheme into a meta-analytic model (i.e. bias-robust model) to obtain less biased estimates of mean effect sizes $\widehat{\beta}$ in the presence of selective reporting. The bias-robust weighting scheme counteracts selective reporting by incorporating the underlying selection mechanisms. Moving to the second step, we treat the bias-robust model as the 'working' model within the CRVE framework ('working' meaning tentatively assumed to be correct; Figure 1). This step helps mitigate potential biases in estimating the standard error estimates $\mathrm{SE}\left(\widehat{\beta}\right)$ that could arise from violating the assumption of statistical independence (i.e. guarding against model misspecification) in the first step bias-robust model. By employing the CRVE, we obtain cluster-robust standard error estimates $\mathrm{SE}\left(\widehat{\beta}\right)$ that ensures the validity of subsequent statistical inferences.

2.2.1 | Step one: Robust point estimate via bias-robust weighting scheme

The first step of our proposed two-step approach aims to mitigate the impact of selective reporting using a bias-robust weighting scheme. Selective reporting can often manifest as a small-study effect (Rosenthal, 1979; Thornton & Lee, 2000), where studies with smaller sample sizes and larger sampling errors tend to report disproportionately large effect sizes (see Section 2.3.2 for another selection mechanism). When this form of selective reporting occurs, a criterion to alleviate its impact on the mean effect size estimate $\widehat{\beta}$ is to assign small studies with small weights. However, the default weighting scheme, inverse variance matrix ($\boldsymbol{\Sigma}_j^{-1}$; see Appendix A for details), is incapable of accomplishing this criterion, particularly when facing large heterogeneity, as it assigns nearly equal weight to each study (Borenstein et al., 2010). In contrast, the inverse sampling variance–covariance can serve as a bias-robust weighting scheme that downweighs smaller studies with low precision, thereby penalizing studies that appear to be 'selectively reported' (large effect size estimates with small sample sizes) (Henmi & Copas, 2010).

We therefore propose using the fixed-effect (FE) model in conjunction with a sampling or within-study variance–covariance matrix (FE+VCV) to serve as a robust point estimator to protect against publication bias. The FE+VCV model is expressed as:

$$y_{ij} = \beta_{\mathrm{FE+VCV}} + e_{ij}, \tag{3}$$

where $\beta_{\mathrm{FE+VCV}}$ denotes the overall effect, y_{ij} denotes the ith effect size estimate in jth study, e_{ij} denotes the sampling error. In jth study, the variance–covariance of the sampling error is given by

$$\mathrm{VCV}_j = \begin{bmatrix} s_{1j}^2 & \cdots & \rho_{1n_j j} s_{1j} s_{n_j j} \\ \vdots & \ddots & \vdots \\ \rho_{n_j 1j} s_{n_j j} s_{1j} & \cdots & s_{n_j j}^2 \end{bmatrix}, \tag{4}$$

VCV_j is also called sampling (with-study) variance–covariance, in which the entries contain information on the precision that can be used to adjust for the impact of selective reporting; note that $\mathrm{VCV} = \mathrm{diag}\left(VCV_1, VCV_2, \ldots, VCV_J\right)$ is often referred to as a block-diagonal matrix where a series of VCV_j are aligned diagonally. Specifically, the sampling variance $\mathrm{Var}(e_{ij}) = s_{ij}^2$ is the diagonal element and the sampling covariance between paired effect size estimates $\mathrm{Cov}(e_{ij}, e_{hj}) = \rho_{ihj} s_{ij} s_{hj}$ is the off-diagonal element (where ρ_{ihj} is the sampling or within-study correlation between ith and hth effect size within jth tudy). In our proposed solution, we assume a constant sampling correlation between each pair of effect sizes within the same study, so that $\rho_j = \rho_{ihj}$ (for relaxing this assumption, see Section 2.2.2).

Notably, three existing meta-analytic models, namely UWLS (Stanley & Doucouliagos, 2015), Henmi-Copas model (Henmi & Copas, 2010), and IVhet models (Doi et al., 2015), also exhibit the capacity to counteract selective reporting (see Appendix A). It turns out that the three are the special cases within the framework we propose, featuring more simplified assumptions. Specifically, these models assume statistically independent effect sizes and sampling errors, characterized by a zero-sampling correlation ρ_{ihj} in the VCV matrix (Equation 4). However, when a study contributes more effect size estimates than another ($n_j > n_k$), these models tend to assign more weights to the study with more effect size estimates. Computationally, when turning the VCV matrix into a weighting matrix, the off-diagonal elements become negative values, which can de-emphasize the studies reporting more effect size estimates when estimating $\hat{\beta}$.

After obtaining $\hat{\beta}$ using the proposed FE model using a within-study VCV matrix (FE+VCV), the following procedures involve performing statistical inferences on the estimated $\hat{\beta}$. One common method for this purpose is the Wald-type test. Alternative methods are also available, such as the likelihood ratio test and permutation test (Viechtbauer et al., 2015). The Wald-type test involves comparing a test statistic t against critical values of a known distribution to test the null hypothesis H_0: $\beta = 0$. The test statistic t is computed using the standard error $\mathrm{SE}(\hat{\beta})$ ($t = \frac{\hat{\beta}}{\mathrm{SE}(\hat{\beta})}$). Under H_0, test statistic t follows (asymptotically) the standard normal distribution or t distribution with $\mathrm{df} = Jn_j - 1$ degrees of freedom (adjustments to df are possible such as $\mathrm{df} = J - 1$ (Hartung, 1999; Knapp & Hartung, 2003; Sidik & Jonkman, 2002)), which can be used to construct a confidence interval and calculate a p-value for the test. However, while the FE+VCV model can reduce the bias of $\hat{\beta}$ with respect to selective reporting, $\mathrm{SE}(\hat{\beta})$ is underestimated due to the failure to account for multiple, statistically dependent effect sizes (effect sizes are arbitrarily correlated within the same study), distorting statistical inference and inflating test statistic t, and Type I error rate (thus reducing p-value).

2.2.2 | Step two: Robust standard error via robust variance estimation

The second step of the proposed two-step approach involves using CRVE to guard against the mis-specified dependence structure in the FE+VCV model from the first step. In Section 2.1, we highlighted that estimating the standard error $\mathrm{SE}(\hat{\beta})$ requires knowledge of the true dependence structure (represented by $\boldsymbol{\Sigma}_j$) for each study in the meta-analysis and deriving the weighting scheme ($\boldsymbol{\Sigma}_j^{-1}$) to minimize $\mathrm{SE}(\hat{\beta})$ (i.e. the efficient estimator). The bias-robust weighting scheme used in the FE+VCV model reduces the bias of β at the expense of biasing its standard error $\mathrm{SE}(\hat{\beta})$. Fortunately, CRVE can provide a remedy by estimating $\mathrm{SE}(\hat{\beta})$ with a dependency structure (imposing mis-specified structures on $\boldsymbol{\Sigma}_j$, including the constant sampling correlation) that is discrepant from the true structure (Hedges et al., 2010; Pustejovsky & Tipton, 2022). In the context of meta-analysis, Hedges and colleagues formalized the use of CRVE to account for misspecification in the dependency structure (Hedges et al., 2010). The elegancy of CRVE is its ability to provide unbiased estimates of standard errors under any weighting scheme, even those significantly different from the fully efficient weighting scheme (as discussed in Section 2.1).

Instead of imposing a pre-specified structure, CRVE empirically approximates the dependency structure ($\boldsymbol{\Sigma}_j$) of each study using products of the residual vector $\hat{\mathbf{e}}_j = \left(e_{1j}, \ldots, e_{ij}, \ldots, e_{n_j j}\right)'$, with the entries $e_{ij} = y_{ij} - \hat{\beta}_{\mathrm{FE+VCV}}$ (Pustejovsky, 2018). Thus, applying CRVE to the FE+VCV model leads to the robust (sandwich) estimator of $\mathrm{Var}(\hat{\beta})$ protecting against a mis-specified dependency structure:

$$\mathrm{Var}\left(\hat{\beta}_{\mathrm{CRVE}}\right) = \left(\sum_{j=1}^{J} \mathbf{1}_j' \mathbf{W}_j \mathbf{1}_j\right)^{-1} \sum_{j=1}^{J} \mathbf{1}_j' \mathbf{W}_j \hat{\mathbf{e}}_j \hat{\mathbf{e}}_j' \mathbf{W}_j \mathbf{1}_j \left(\sum_{j=1}^{J} \mathbf{1}_j' \mathbf{W}_j \mathbf{1}_j\right)^{-1}, \quad (5)$$

When the number of studies J is sufficiently large, the average of the residual products of all J studies can accurately approximate the dependency structure. Therefore, $\mathrm{Var}(\hat{\beta}_{\mathrm{CRVE}})$ converges to the true sampling variance of the overall effect as J becomes infinite (Hedges et al., 2010). As a result, statistical inferences based on the cluster-robust standard error $\mathrm{SE}(\hat{\beta}_{\mathrm{CRVE}}) = \sqrt{\mathrm{Var}(\hat{\beta}_{\mathrm{CRVE}})}$ are statistically valid. Importantly, the cluster-robust standard errors $\mathrm{SE}(\hat{\beta}_{\mathrm{CRVE}})$ can ensure a valid null-hypothesis test of the overall effect $\hat{\beta}$ and correct construction of the confidence intervals. We note that the cluster-robust standard error $\mathrm{SE}(\hat{\beta}_{\mathrm{CRVE}})$ of the original CRVE (Hedges et al., 2010) is downwardly biased and the corresponding hypothesis tests do not adequately control Type I error rates unless the number of independent studies is large. Fortunately, CRVE estimators that incorporate a small-sample correction have been proposed to improve their accuracy (Pustejovsky & Tipton, 2018). These small-sample CRVE estimators include using adjustment matrices (e.g. CR1, CR2 and CR3), robust-wild bootstrapping techniques and adjusting the degrees of freedom for statistical tests of model coefficients (Joshi et al., 2022; Pustejovsky & Tipton, 2018; Tipton, 2015).

2.3 | Performance assessment

We compared our method against a multilevel version of PET-PEESE by assessing the bias of the overall effect estimate in the presence of selective reporting and statistical dependence. Then, we assessed systematic discrepancies in meta-analytic estimates between our proposed approach and the meta-analytic method that ignores selective reporting and statistical dependence. Below, we briefly describe (1) the large dataset of $N = 448$ meta-analyses that we used for empirical comparisons, (2) the benchmark meta-analytic model commonly used by ecologists and evolutionary biologists, (3) two selective reporting detection methods with different underlying selection mechanisms used to diagnose selective reporting in the $N=448$ meta-analyses, and (4) how to obtain the 'proxies' for the true effects of the 448 meta-analyses using PET-PEESE.

2.3.1 | Dataset compilation

The dataset used in our study consisted of $N=448$ meta-analyses that were gathered by Costello and Fox (2022). All meta-analyses included claimed adherence to the PRISMA reporting guidelines (Moher et al., 2009). We further performed data cleaning to suit our analysis. Specifically, we eliminated cases with zero sampling variance, classified effect size measures into four categories (standardized mean difference [SMD] family [i.e. Cohen's *d* and Hedge's *g*], log response ratio [lnRR], Fisher's *Zr* and uncommon measures [i.e. mean difference, regression slope, risk ratio and odds ratio]) (Nakagawa & Cuthill, 2007). We dropped meta-analysis datasets that had convergence issues during model fitting, despite trying different numerical optimizers and optimization specifications in an attempt to rectify the problems (i.e. number of iterations, step size and threshold). After the cleaning process, 448 meta-analysis datasets were included (Yang, 2024). All 488 meta-analyses reported multiple, statistically dependent effect sizes, with each study contributing eight effect size estimates on average, which indicates effect sizes were statistically dependent and did not contribute unique information (intra-class correlation among the true effect sizes was ICC$=0.52$).

2.3.2 | Standard (benchmark) method: Multilevel meta-analytic (MLMA) model

The multilevel meta-analytic (MLMA) model is a commonly used method for handling dependent effect sizes (Nakagawa & Santos, 2012). The basic MLMA model is a three-level meta-analytic model that includes random-effects at the between-study and within-study levels. It can be expressed as:

$$y_{ij}=\beta_{\mathrm{MLMA}}+u_{(b)j}+u_{(w)ij}+e_{ij}, \tag{6}$$

where β_{MLMA} denotes the model intercept, representing the overall effect or mean effect size; $u_{(b)j}$ is a random-effects at between-study level with $\mathrm{Var}\left(u_{(b)j}\right)=\tau_b^2$, which captures between-study heterogeneity; $u_{(w)ij}$ is a random-effects at the within-study level with $\mathrm{Var}\left(u_{(w)ij}\right)=\tau_w^2$, which captures within-study heterogeneity; e_{ij} is the corresponding sampling error, with $\mathrm{Var}\left(e_{ij}\right)=s_{ij}^2$ and $\mathrm{Cov}\left(e_{ij},e_{hj}\right)=0$. The estimates of β and $\mathrm{SE}\left(\hat{\beta}\right)$ can be reached using Equations (1) and (2), with weights equal to inverse variance–covariances. The weights for MLMA for each study are written as (Viechtbauer, 2007):

$$\mathbf{W}_{j\mathrm{MLMA}}=\begin{bmatrix} \hat{\tau}_b^2+\hat{\tau}_w^2+s_{1j}^2 & \cdots & \hat{\tau}_b^2 \\ \vdots & \ddots & \vdots \\ \hat{\tau}_b^2 & \cdots & \hat{\tau}_b^2+\hat{\tau}_w^2+s_{n_jj}^2 \end{bmatrix}^{-1}, \tag{7}$$

The MLMA model offers several advantages, including readily available implementation software and not requiring sampling correlations (Equation 4), which have made it a standard benchmark method for dealing with dependent effect sizes in many disciplines (Nakagawa & Santos, 2012; Nakagawa, Yang, et al., 2023; Van den Noortgate et al., 2013; Yang et al., 2022). However, one unappreciated limitation of MLMA is its inverse variance weighting scheme, which can lead to an overestimation of β when selective reporting is present (see Section 2.1).

2.3.3 | Detecting selective reporting using multilevel Egger's test and ad hoc selection model

Detecting selective reporting in the presence of dependent effect sizes poses a challenge, particularly in our dataset of 448 meta-analyses, as most methods violate the assumption of statistical independence (Nakagawa et al., 2022). We employed the multilevel Egger's test (MLMA Egger) as our first method. Extensive simulations have shown that it maintains close-to-nominal Type I error rates in the presence of dependent effect sizes (Fernández-Castilla et al., 2021; Rodgers & Pustejovsky, 2021). The MLMA Egger is obtained by adding the sampling error s_{ij} (square root of sampling error s_{ij}^2) as a predictor into MLMA model (Equation 7):

$$y_{ij}=\beta_{PET}+\beta_1 s_{ij}+u_{(b)j}+u_{(w)ij}+e_{ij}, \tag{8}$$

A statistically significant coefficient β_1 implies the presence of selective reporting. Therefore, MLMA Egger can detect selective reporting arising from a selection mechanism where larger effect sizes, albeit with small precision (large sampling error), are more likely to be published (see limitations in Section 5).

The second method employed in our study was the selection or weight function model. Specifically, we used the ad hoc three-parameter selection model (3PSM) (Rodgers & Pustejovsky, 2021; Vevea & Hedges, 1995). The ad hoc 3PSM involves four procedures: (1) using ad hoc strategies to account for statistical dependence, (2) modelling the data-generation process, (3) modelling the selection process and (4) employing a likelihood ratio test (LRT) to detect selective reporting. In the first procedure, we aggregated effect size estimates and sampling variances, assuming a constant sampling correlation of 0.5 (Noble et al., 2017; Yang et al., 2022). For the second procedure, we used a random-effects model:

$$\bar{y}_j=\beta+u_j+\bar{e}_j, \tag{9}$$

where $\bar{y}_j$ represents the aggregated effect size estimates for jth study, $\bar{e}_j$ is the sampling error, with $\mathrm{Var}\left(\bar{e}_j\right)=\bar{s}_j^2$ where $\bar{s}_j^2$ is the aggregated sampling variance corresponding to $\bar{y}_j$. The third procedure involved calculating the probability that effect size estimates are published:

$$\Psi=\frac{\pi^{p\geq\alpha}}{\pi^{p<\alpha}}, \tag{10}$$

where Ψ is the probability of publishing a non-significant effect size at α level ($\pi^{p\geq\alpha}$) relative to the probability of publishing a significant one ($\pi^{p<\alpha}$). In the fourth procedure, we used an LRT to assess the null hypothesis ($\Psi = 1$), indicating no selection of significant effect sizes. While ad hoc 3PSM has slightly inflated Type I error rates (Rodgers & Pustejovsky, 2021), we employed it as a sensitivity analysis for diagnosing selective reporting. This approach can detect selective reporting arising from the selection mechanism determining how the probability of publishing an effect size depends on its statistical significance (Equation 10).

2.3.4 | Estimating the proxy for the true effect using PET-PEESE

To evaluate the empirical performance of our proposed method, we conducted a comparison with publication-bias correction methods designed to approximate true effect sizes. PET-PEESE shows generally less bias and overestimation compared to other bias adjustment methods (Kvarven et al., 2020; Stanley & Doucouliagos, 2014). However, the original PET-PEESE can only be used to correct for publication bias for statistically dependent effect sizes. For our study, we employed the multilevel version of PET-PEESE to address statistically dependent effect sizes (Nakagawa et al., 2022). The multilevel PET-PEESE method comprises two main procedures. The first, PET (precision effect test), involves a linear meta-regression that approximates the truncated selection model (Stanley & Doucouliagos, 2014). The PET is equivalent to Equation (8) with β replaced by β_{PET}. In other words, the PET approach uses the sampling standard error as a moderator. The second procedure, PEESE (precision effect estimate with standard error), represents a quadratic meta-regression approximation to the truncated selection model and can be expressed as

$$y_{ij} = \beta_{\mathrm{PEESE}} + \beta_1 s_{ij}^2 + u_{(b)j} + u_{(w)ij} + e_{ij}, \tag{11}$$

When the precision measure is infinitely large ($s_{ij} = s_{ij}^2 = 0$), the intercept (β_{PET} and β_{PEESE}) provides an approximation of the true effect β_{true}. We used β_{PET} as the proxy for β_{true} when it was not significant at the $a = 0.1$ level, and β_{PEESE} as the proxy for β_{true} when it was significant at the $a = 0.1$ level (Stanley & Doucouliagos, 2014). While PET-PEESE provides a mean effect size estimate close to the true effect when the number of studies is above 10 (Stanley & Doucouliagos, 2014), the statistical inference is notably conservative due to high variability around the estimate (see details in Section 5).

2.3.5 | Performance metric

We employed three sets of metrics to evaluate the performance of the proposed approach. The first set comprised standard performance metrics, including mean error (ME), mean square error (MSE), and overestimation factor (OF). ME quantifies the average deviation of our approach's overall effect estimate from the 'true' effect, approximated by the multilevel PET-PEESE (a proxy for the 'true' effect $\hat{\beta}_{\mathrm{true}}$; see Section 2.3.4). ME is commonly known as bias ($\frac{1}{N}\sum_1^N \hat{\beta}_{\mathrm{FE+VCV}} - \hat{\beta}_{\mathrm{true}}$), where N denotes the number of meta-analytic datasets. MSE gauges efficiency, considering both the average error and variance ($\frac{1}{N}\sum_1^N \left(\hat{\beta}_{\mathrm{FE+VCV}} - \hat{\beta}_{\mathrm{true}}\right)^2$). OF represents the ratio of our approach to the proxy for the true effect ($\frac{1}{N}\left(\frac{\hat{\beta}_{\mathrm{FE+VCV}}}{\hat{\beta}_{\mathrm{true}}}\right)$). Given our expectation that our proposed method would perform comparably to multilevel PET-PEESE, these metrics were instrumental in testing their equivalence in estimating the overall effect with selective reporting. We also assessed the performance of UWLS, an existing meta-analytic model robust to selective reporting but neglecting statistical dependence (see Appendix A). We anticipated the potential overestimation of UWLS in the presence of multiple, statistically dependent effect sizes compared to our proposed method (see Section 2.2.1).

The second set of performance metrics included the ln-transformed ratio of the overall effect estimates $\hat{\beta}$ obtained from the MLMA model to those from the first step (i.e. FE+VCV model) of our proposed approach (Lajeunesse, 2011; Pustejovsky, 2018). We used it to evaluate inflation of the overall effect estimates arising from the use of the MLMA model in the presence of selective reporting. The third set of performance metrics included the log-transformed ratio of $\mathrm{SE}\left(\hat{\beta}\right)$ (the square-root of $\mathrm{Var}\left(\hat{\beta}\right)$) obtained from the MLMA model to those from the second step (applying CRVE to fitted model) of our proposed approach (Nakagawa et al., 2015). As the sign of the effect size estimate does not provide information for assessing inflation magnitude, we computed the folded mean and sampling variance of the second and third sets of metrics using the folded normal distribution (Morrissey, 2016; Yang et al., 2023). Subsequently, we employed a random-effect meta-analysis to quantify the overall inflation across $N = 448$ meta-analyses. This comprehensive evaluation strategy allowed us to robustly gauge the performance and comparative effectiveness of our proposed two-step approach.

3 | RESULTS

3.1 | Comparisons between the proposed two-step method and multilevel PET-PEESE

Table 1 and Figure 2 report the mean error (ME), mean square error (MSE), and overestimation factor (OF) for our proposed approach (FE+VCV+CRVE), assuming the overall effect estimates $\hat{\beta}$ from the multilevel PET-PEESE adjustment method is the proxy of the true effects. The results from Table 1 and Figure 2 indicate that our proposed approach is comparable to the multilevel PET-PEESE in terms of estimating the overall effects $\hat{\beta}$, especially when the number of studies N is above 10 (Table 1; Figure 2). For example, under two selective reporting processes, as identified by MLMA Egger and ad hoc 3PSM, our proposed approach achieved a low ME (0.047 and 0.020). In the absence of selective reporting, the ME was negligible (Figure 2). However, UWLS, a model robust to publication bias assuming statistically independent effect sizes, demonstrated upward bias with a mean error (ME) of 0.227 in the presence of selective reporting (Table 1; Figure 2). These results justify our use

Selective reporting	Detection method	Estimator	ME	MSE	OF
Presence	MLMA Egger	FE+VCV+CRVE	0.047	0.268	1.2
		UWLS+CRVE	0.227	0.272	2.0
	Ad hoc 3PSM	FE+VCV+CRVE	0.020	0.330	1.1
		UWLS+CRVE	0.147	0.268	1.5
Absence	MLMA Egger	FE+VCV+CRVE	−0.016	0.397	1.0
		UWLS+CRVE	−0.019	0.379	1.0
	Ad hoc 3PSM	FE+VCV+CRVE	0.004	0.387	1.0
		UWLS+CRVE	0.042	0.385	1.1

TABLE 1 Results for different performance metrics used to assess the equivalence between the proposed method and the multilevel PET-PEESE.

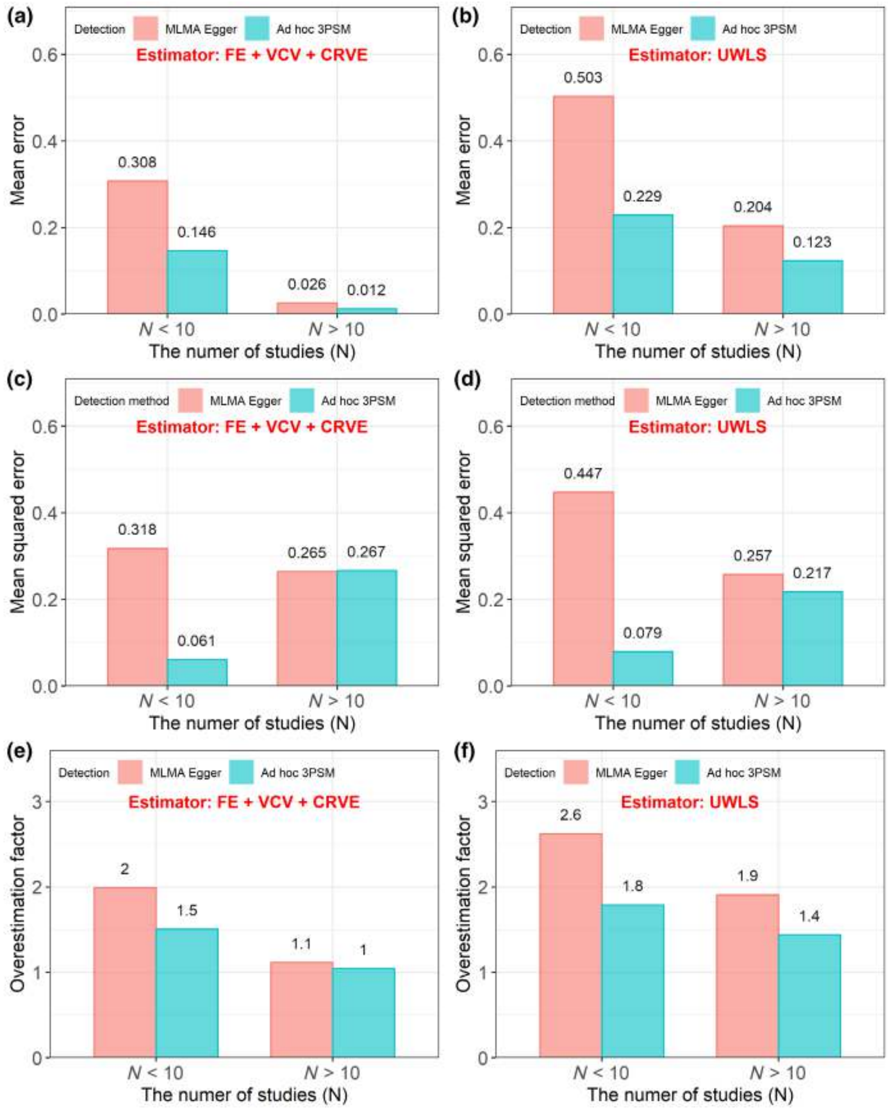

FIGURE 2 Comparing the proposed two-step method against the publication-bias adjustment method stratified by the number of studies (N). The latter, multilevel PET-PEESE, provides an estimate of the overall effect $\hat{\beta}$ that are 'corrected' for the selection process, representing true effects before selective reporting. Our proposed method employs a fixed-effect model with a within-study variance–covariance matrix (FE+VCV) as the first step to estimate the overall effect $\hat{\beta}$ that are robust to the selection process. In the second step, it uses cluster robust variance estimation (CRVE) to estimate standard errors that are robust to multiple, statistically dependent effect sizes. The results for the mean error (ME), mean square error (MSE) and overestimation factor (OF) indicate that our proposed method performs comparably to multilevel PET-PEESE in terms of estimation of average population effect (panels a, c and e). In contrast, while unrestricted weighted least square (UWLS) estimator can counteract selective reporting for statistically independent effect sizes, it overestimates the overall effect $\hat{\beta}$ in the presence of multiple, statistically dependent effect sizes (panels b, d and f). Two types of methods are used to detect selective reporting (Rodgers & Pustejovsky, 2021): multilevel Egger's test (MLMA Egger) and ad hoc three-parameter selection model (ad hoc 3PMS). The values of performance metrics corresponding to different estimators are shown at the top of the bars.

of the proposed two-step method compared to the baseline model (MLMA) in the next two sections.

3.2 | Inflation of the mean effect size if ignoring selective reporting

In the presence of selective reporting, as identified by MLMA Egger, the overall effects $\hat{\beta}$ estimated by the benchmark method (MLMA model), consistently exceeded those obtained from the first step bias-robust model of our proposed method (FE+VCV; Figure 3). Specifically, out of the 149 meta-analyses with selective reporting, 141 exhibited an overestimated $\hat{\beta}$ when using the MLMA model (Figure 4), indicating 95% of meta-analyses exaggerated their mean effects if not employing any correction for selective reporting. On average, $\hat{\beta}$ derived from the MLMA model was 114.4% (0.763, 95% CI=[0.690, 0.835]; Table S1) larger than that from the FE+VCV model (Figure 3). In contrast, for the meta-analyses without selective

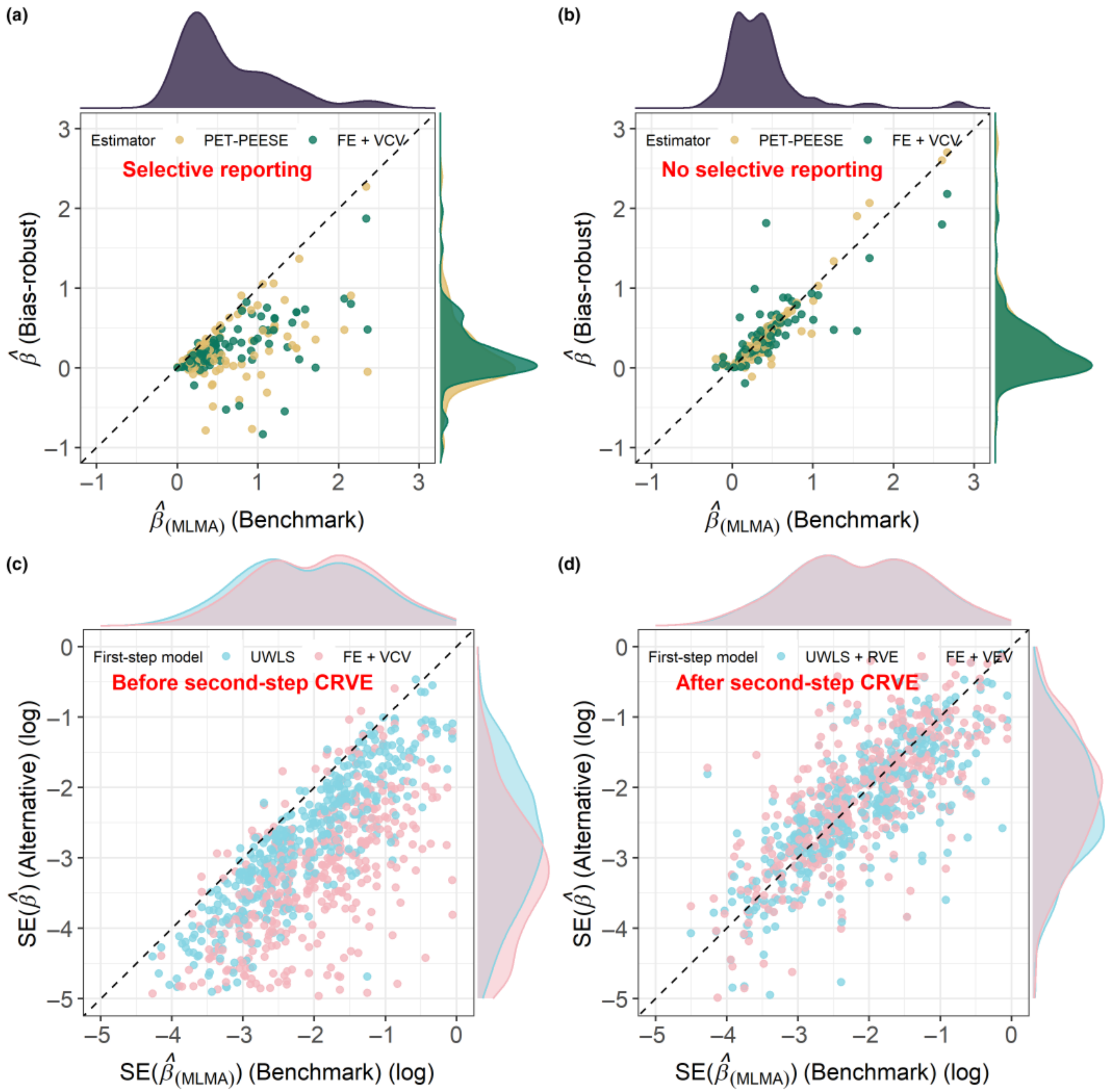

FIGURE 3 Comparison of the overall effect estimates $\hat{\beta}$ and their standard errors $\mathrm{SE}\left(\hat{\beta}\right)$ obtained from different models. The dashed diagonal line is $y = x$, indicating where $\hat{\beta}$ and $\mathrm{SE}\left(\hat{\beta}\right)$ would fall if there were no discrepancies between the models. The benchmark method, MLMA (multilevel meta-analysis model), is considered a standard method for meta-analysis in the presence of statistical dependence. The first-step bias-robust model of the proposed approach (FE+VCV model) and PET-PEESE adjustment method consistently mitigated the impact of selective reporting (a and b). Cluster robust variance estimation (CRVE), the second step of our approach, provided consistent estimates of (robust-robust) standard error $\mathrm{SE}\left(\hat{\beta}\right)$ for subsequent statistical inference (c and d). For other details, see Figure 2.

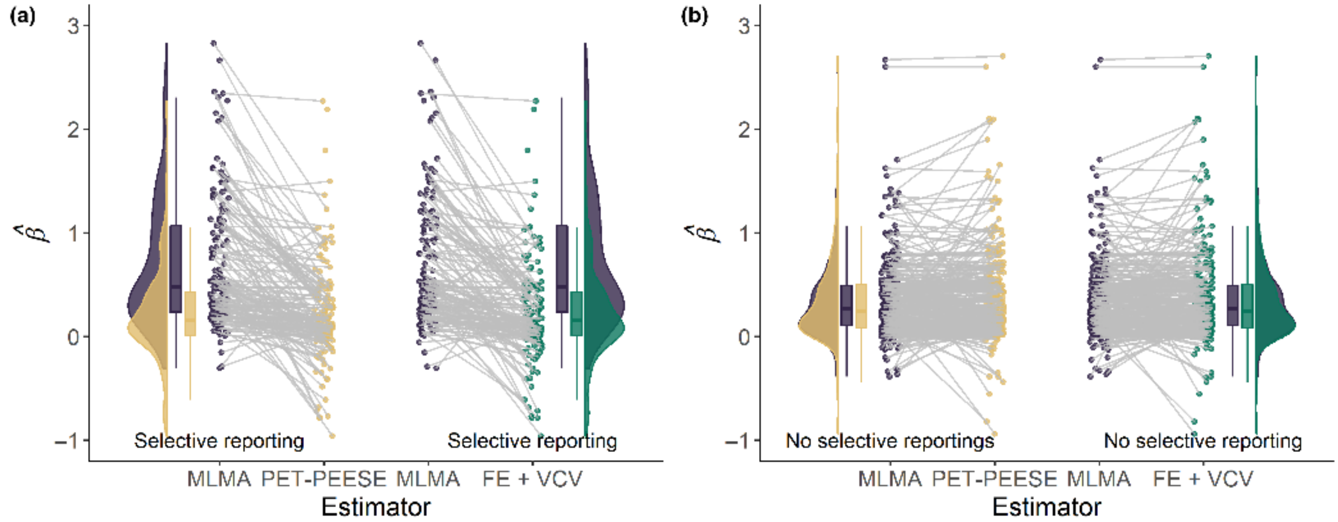

FIGURE 4 Paired comparison of the estimated overall effects $\hat{\beta}$ obtained from the benchmark method (MLMA), the first-step bias-robust model of the proposed approach (FE + VCV) and PET-PEESE adjustment method. For other details, refer to the legend in Figure 3. The comparison was in the context of meta-analysis with selective reporting (b) and without selective reporting (a).

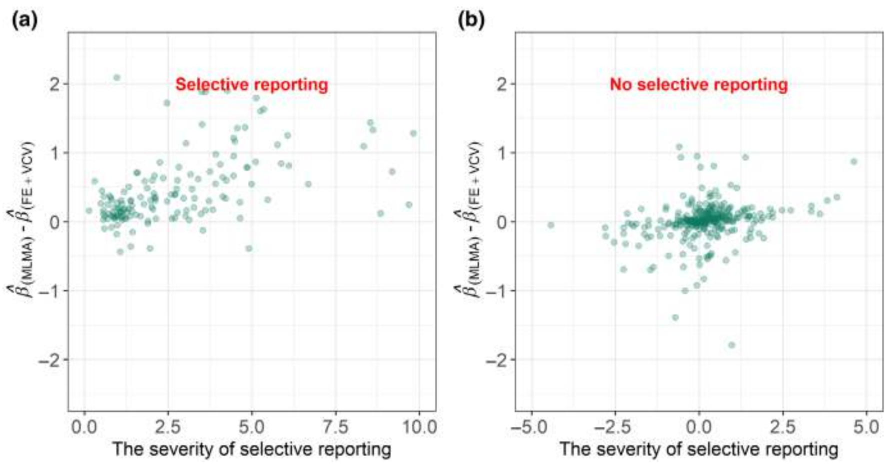

FIGURE 5 The relationship between the difference in the overall effect estimates $\hat{\beta}$ and the severity of publication bias. The slope of the extended Egger test represents the severity of publication bias. The greater the slope, the greater the bias in the overall effect estimate (a), which would not have occurred in the absence of selective reporting (b).

reporting, there were no systematic differences in the estimated $\hat{\beta}$ among different models. Importantly, the magnitude of the discrepancy in the $\hat{\beta}$ between the models was positively associated with the severity of selective reporting (0.039, 95% CI = [0.001, 0.077]; Table S2; Figure 5). Selective reporting at a significance level of $\alpha = 0.05$ showed similar patterns with a nominal significance level ($\alpha = 0.05$) (see Figures S1–S3). Similar patterns were also observed for selective reporting detected by ad hoc 3PSM (Figures S1–S5), which assumes a distinct selection mechanism from Egger's test.

3.3 | Underestimation of the standard error if ignoring statistical dependence

Before applying the second step of the proposed method, the FE + VCV model consistently underestimated the $\mathrm{SE}\left(\hat{\beta}\right)$ analysing meta-analyses with statistical dependence (Figure 2c). Among the 448 published meta-analyses analysed, 428 showed an underestimation of $\mathrm{SE}\left(\hat{\beta}\right)$ (Figure 3c), indicating that the statistical inference of 96% of the published meta-analyses would be distorted.

On average, the $\mathrm{SE}\left(\hat{\beta}\right)$ derived from the FE + VCV model were 382.3% (1.573, 95% CI = [1.469, 1.678]) smaller, than that obtained from the MLMA model when considering datasets with multiple, statistically dependent effect sizes (Figure 3c; Table S3). After applying the second step of our proposed method, CRVE, no systematic difference was found in the estimated $\mathrm{SE}\left(\hat{\beta}\right)$ (Figure 3d). There was no statistically significant association observed between the magnitude of the discrepancy in the estimated $\mathrm{SE}\left(\hat{\beta}\right)$ between the FE + VCV model and the MLMA model and the degree of statistical dependence (0.152, 95% CI = [−0.067, 0.371] and −0.108, 95% CI = [−0.445, 0.230], respectively; Figure 6).

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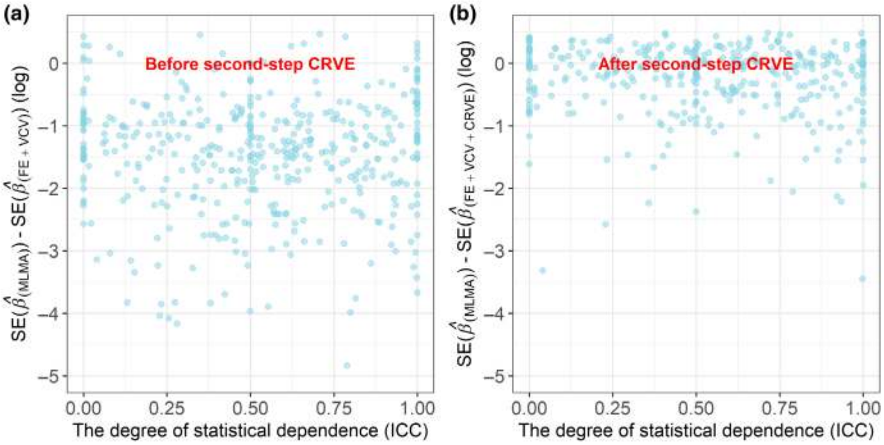

FIGURE 6 The relationship between the difference in the estimated standard error of the mean effect sizes $SE\left(\hat{\beta}\right)$ and the indicator of the degree of statistical dependence. The intraclass correlation coefficient (ICC) represents the degree of dependence among effect sizes in a meta-analysis. For other details, refer to the legends in Figures 3 and 5. Failure to account for statistical dependence leads to biased standard errors (a), and the use of clustered variance estimates provides an effective remedy (b).

4 | IMPLEMENTATION AND VISUALIZATION: AN ILLUSTRATIVE EXAMPLE

We created a tutorial to allow practitioners to apply our proposed approach. We selected a published meta-analysis that claimed the existence of publication bias. This example meta-analysis examined the effect of herbivore interaction on fitness based on 179 species, 167 studies and 1640 effect sizes (Bird et al., 2019). In line with the publication bias test conducted in the original publication, our re-analysis also confirmed evidence for publication bias (0.876, 95% CI = [0.623, 1.128], see Figure S5 in the tutorial). To address the statistical dependence among effect sizes (with 10 effect sizes/study), the original publication employed Bayesian multilevel meta-analytic modelling with phylogenetic relatedness, study and observation identities as random effects.

To implement the first step of the proposed method, we first used the vcalc() function in the metafor package (Viechtbauer, 2010) to construct a sampling VCV matrix (Figure 6), assuming a constant sampling correlation of 0.5 (Noble et al., 2017). The sensitivity analysis was conducted to examine the extent to which the mean effect was sensitive to the assumption of within-study (sampling) correlation ρ_{ihj} values used for constructing VCV matrix (see tutorial). Then, we used the rma.mv() function to fit a fixed-effect model using the constructed VCV matrix (FE + VCV model), obtaining the publication-bias-robust population mean effect of herbivore interaction. In the second step, we used the coef_test() in metafor (or coef_test() in the clubSandwich (Pustejovsky & Tipton, 2022)) to address the misspecified dependence structure in the fitted FE + VCV model. This allowed us to compute robust standard errors and perform statistical inference on the interaction between herbivores. The result using our proposed approach indicated a minimal interaction between herbivores ($\hat{\beta} = 0.075$, $SE\left(\hat{\beta}\right) = 0.054$, 95% CI = [−0.034, 0.184], $t_{52} = 1.375$, p-value = 0.175; Figure 7). In contrast, the original publication reported a statistically significant interaction between herbivores ($\hat{\beta}$ = 0.275, $SE\left(\hat{\beta}\right) = 0.047$, 95% CI = [0.181, 0.368], $t_{166} = 5.783$, p-value <0.001). We developed the pub_bias_plot() function, which can be used in conjunction with the orchard_plot()function from the orchaRd package (Nakagawa et al., 2021). Their combined use provides a new graphical plot for visualizing the impact of publication bias on parameter estimation and inference (Figure 7), allowing for a visual assessment of the robustness of the meta-analytic findings and facilitating transparent reporting.

5 | DISCUSSION

To correctly estimate overall meta-analytic means and draw valid inferences in the presence of selective reporting and statistical dependence, we propose a readily implementable two-step approach. Under two selective reporting processes identified by Egger's test and the selection model, the fixed-effect (FE) model in conjunction with a sampling variance–covariance matrix (FE + VCV) employed in the first step demonstrated comparable performance to the publication-bias adjustment method (multilevel PET-PEESE). Our method yielded less biased overall effects compared to the standard multilevel meta-analysis model (MLMA; Figures 2 and 3). The use of MLMA resulted in an average overestimation of the overall effects by 110% when selective reporting was present (Figure 4). Furthermore, the severity of selective reporting was positively associated with the discrepancy in overall effect estimates (Figure 5), although the observed positive correlation might be an artefact that arise from using the extended Egger's test to detect publication bias. Including cluster-robust variance estimation (CRVE) as the second step successfully addressed the issue of statistical dependence and achieved comparable estimates of standard errors across the models (Figure 3). On average, the CRVE corrected the estimates of standard errors by 120%. If these underestimated standard errors were to be used for statistical inference, it would lead to artefactually narrower CIs, increasing Type I error (false positive) rates in meta-analytic

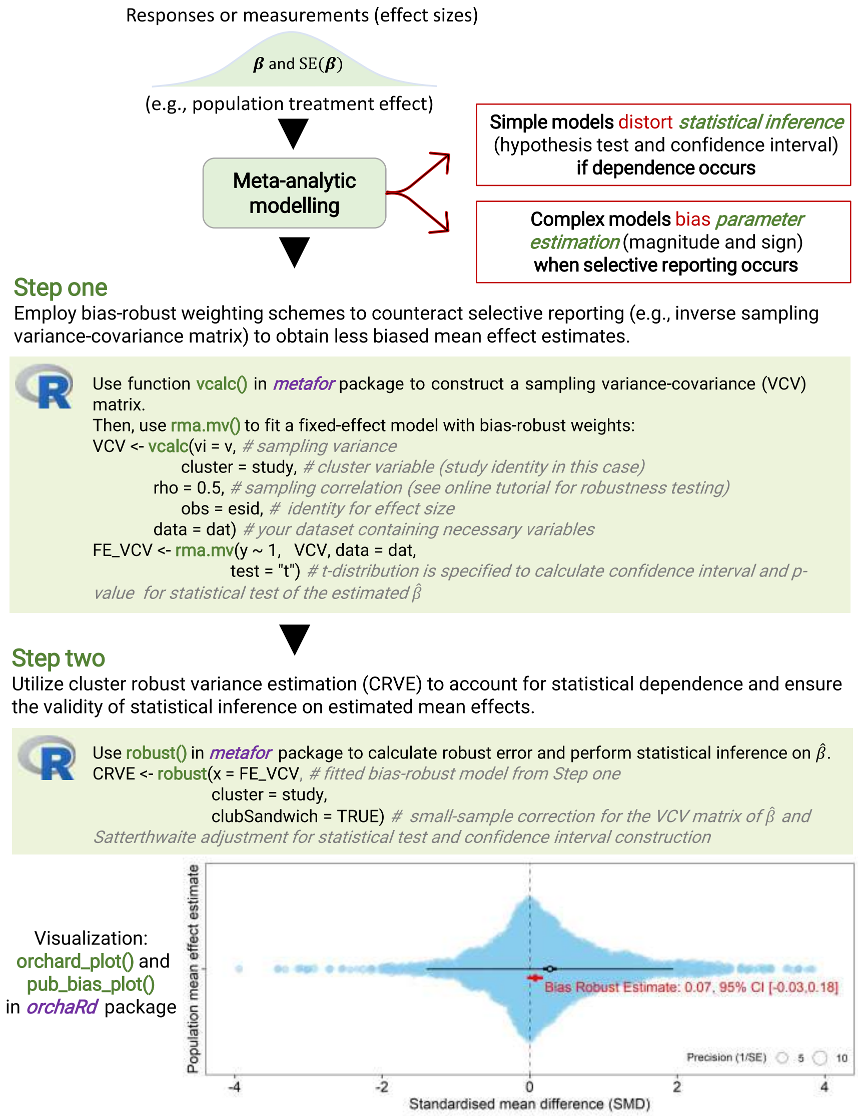

FIGURE 7 The implementation of the proposed two-step meta-analytic modelling approach, along with a novel visualization, using open-source software. The metafor (Viechtbauer, 2010) package is utilized for building two-step models, while the orchaRd (Nakagawa et al., 2021) package is employed for graphical representation. For a detailed step-by-step tutorial demonstrating two alternative first-step models, please refer to the webpage. The bottom graph provides a visual solution to show publication-bias-robust parameter estimation and inference derived from the proposed two-step approach, serving as a sensitivity analysis. It also includes essential elements for transparent reporting, such as 95% confidence intervals (CIs), 95% prediction intervals (PIs) and precision (inverse standard error).

results. Below, we discuss the innovations, extensions, limitations, and future perspectives of the proposed method.

5.1 | Extending the tools for ascertaining the impacts of publication bias in meta-analysis

Despite dedicated methods being available for adjusting selective reporting, including the fail-safe-*N* method, trim-and-fill, selection models and *p*-curve methods, these are not easily applied to complex hierarchical data. Failing to address statistical dependence (Fernández-Castilla et al., 2021) can result in unreliable publication-bias-corrected mean effect size estimates. A recently proposed multilevel version of Egger's regression (Nakagawa et al., 2022; Rodgers & Pustejovsky, 2021) shows promise in addressing publication bias with complex hierarchical data by regressing effect size estimates against their sampling variances, also known as PEESE (Stanley & Doucouliagos, 2014), while controlling for statistical dependence. The intercept in the PET-PEESE model could be interpreted as the publication-biased-adjusted mean effect, assuming an infinitely large sample size. The publication-biased-adjusted mean effect represents a marginalized mean in the context of a meta-regression model, after accounting for impacts from informative predictors under infinite precision (or sample size). Unfortunately, informative predictors are rarely fully identified in published studies (Yang et al., 2023). Additionally, extrapolation involved in marginalizing predictor effects may yield poor estimates of publication-biased-adjusted effect size and could also affect the magnitude of standard errors (Figure 6), influencing statistical inference. In contrast, the proposed two-step method does not rely on extrapolation and relaxes the assumption of infinite precision and information from predictors. Our proposed approach should be used as an effective sensitivity analysis to understand the effects of publication bias on the inferences drawn from a meta-analysis.

5.2 | A general framework for incorporating weighting schemes

Our approach offers an effective way of addressing selective reporting by separating weighting schemes for point estimates from the estimation of robust standard errors. While the default inverse variance weighting scheme provides the best linear unbiased estimator of model coefficients only for representative data, it causes bias to the point estimate of the mean population effect in the presence of selective reporting. In contrast, the inverse sampling VCV weighting schemes used in the first step have prioritized reducing the bias of mean effects when the data were not representative (as in the case of selective reporting). Technically, the FE+VCV approach borrows from principles established under multivariate models but does not require researchers to fit random-effects structures (Yang et al., 2022). The CRVE separates the choice of weighting scheme from the estimation of standard errors and uses the residual distribution to approximate the true error distribution. Statistical inferences relying on robust standard errors remain valid under any weighting schemes (Hedges et al., 2010; Pustejovsky & Tipton, 2018, 2022; Tanner-Smith & Tipton, 2014; Tipton, 2015), if the number of independent studies is large. This is particularly relevant when using bias-robust weighting schemes, as in our proposed approach, where the assumed error distribution deviates from the true distribution.

More broadly, our method of tailoring and reconceptualising weighting can be extended to deal with three major sources of bias pertaining to meta-analytic weights. The first source of bias stems from questionable research practices. This includes selective reporting, such as publication bias, which has been addressed in the present study. Of relevance, poor quality studies or studies with a risk of bias can be potentially handled by using meta-analytic quality models within the CRVE framework (Doi & Thalib, 2008), where study quality information is incorporated into the weighting schemes.

The second type of bias arises from using small sample sizes in primary studies. The proposed two-step approach can help ameliorate the impacts of small sample sizes on the estimation of sampling variances for various effect size measures, which, in turn, increases the accuracy of the weights converted from sampling variances. Current formulas used to compute sampling variance for common effect size measures (e.g. SMD, log response ratio, Fisher's *Zr*) are derived under the assumption of large sample sizes (Borenstein et al., 2021). To mitigate this issue, one straightforward solution is to average the effect size estimates across all included studies and use this average to calculate the sampling variance (the so-called smooth estimator) (Doncaster & Spake, 2018). As the number of studies increases, the averaged effect size converges to the true underlying effect. This improvement in handling small sample sizes also offers two additional benefits. It allows for the inclusion of primary studies with missing standard deviations (Doncaster & Spake, 2018) and enables data imputation for those missing standard deviations (Nakagawa, Noble, et al., 2023).

The third type of bias originates from the statistical properties of certain effect size measures, where the point estimate intrinsically correlates with its sampling variance. This is particularly relevant for effect size measures such as SMD, partial correlation coefficient and log odds ratio (Bakbergenuly et al., 2020; Nakagawa et al., 2022). To address this inherent correlation and mitigate bias, effective-sample-size-based or unit-weighting schemes have been proposed (Bakbergenuly et al., 2020). However, these weighting schemes are not easily extendable to the framework of multilevel models, which are essential for accounting for statistical dependence and avoiding inflated Type I error rates.

5.3 | Limitations and future opportunities

Two potential limitations should be noted about our proposed two-step approach for dealing with publication biases. First, the diagnostic methods used to identify selective reporting, namely the multilevel version of Egger's test and the ad hoc three-parameter selection model, each have their own weaknesses. The multilevel version of Egger's test may have limited statistical power under certain circumstances (Nakagawa et al., 2022; Rodgers & Pustejovsky, 2021), while the ad hoc three-parameter selection model exhibits inflated Type I error rates (high false positives) in detecting selective reporting. However, the convergence of results from both methods lends support to the effectiveness of our proposed approach in handling selective reporting. Additionally, we conducted a sensitivity analysis by relaxing the significance level to 0.1. The results from the significance level of 0.1 aligned with that from the nominal level of 0.05 (Figures S1–S3).

Second, we did not employ a simulation approach for assessing the empirical performance of statistical models, as our focus was to establish a sensitivity analysis. From a realistic perspective, there is limited quantitative knowledge available regarding the dependence structure and patterns of publication bias in real-world scenarios (Bramley et al., 2021). Designing simulations that accurately reflect these characteristics can be challenging. For example, while previous simulations suggested high false positives for the ad hoc three-parameter selection model (Rodgers & Pustejovsky, 2021), our datasets demonstrated a different result, with only 70 cases of selective reporting identified by this model compared to 149 cases identified by the multilevel version of Egger's regression. Therefore, our study leveraged the richness of published meta-analyses, which are more likely to capture the diverse range of dependency structures and publication bias patterns in real-world settings (see Figure 1). An extensive simulation comparing all models that are robust against selective reporting would still be valuable in the future (Yang et al., in preparation). Additionally, our study specifically focused on the intercept-only meta-analysis models. While meta-regression models with categorical predictors can be transformed into subgroup intercept-only models, further investigation through simulations is needed to assess the generalizability of the proposed method to meta-regression models.

6 | CONCLUSIONS

While sophisticated multi-level meta-analytic models are essential when no selective reporting is present, it inflates the effect size estimation and may lead to misleading conclusions. The field of meta-analysis has strived to use sophisticated model structures to correctly capture the underlying data-generation process. In contrast, our proposed two-step approach takes a different perspective by prioritizing the adjustment for selective reporting in the first step and ensuring the validity of subsequent statistical inference by CRVE in the second step. This shift in focus emphasizes the development of appropriate weighting strategies to reduce bias in meta-analytic evidence when the data are non-random.

We emphasize that the proposed approach is not intended to replace standard meta-analytic models (i.e. MLMA). More specifically, the proposed approach is not able to provide a heterogeneity estimate and its performance needs to be validated through simulation. Importantly, the performance of different models is contingent upon the true data-generating mechanisms which are rarely known. Therefore, we expect that the proposed approach serves as a sensitivity analysis to standard methods when interested in the mean population effect.

In alignment with the move towards a 'multiverse' analytical workflow (Steegen et al., 2016), we advocate for the routine use of our two-step method and its associated graphical tool as a sensitivity analysis (Figure 7). Complex hierarchical dependency structures and publication biases are typical of meta-analyses. As such, it is becoming increasingly critical to explore and present multiple plausible analyses instead of relying solely on a single model (i.e. multiverse meta-analytic modelling) (Wagenmakers et al., 2022). A more detailed assessment of the robustness of meta-analytic models would improve transparency and could be used to strengthen the meta-analytic evidence, necessary to build the quantitative evidence that underpins meta-analytic research and decision-making.

AUTHOR CONTRIBUTIONS

Yefeng Yang and Shinichi Nakagawa conceptualized the idea and drafted the manuscript. Yefeng Yang analysed the data with the help of Shinichi Nakagawa. Yefeng Yang led the creation of the accompanying webpage working with Coralie Williams and Shinichi Nakagawa. Yefeng Yang led all the visualizations with the help of Malgorzata Lagisz. Daniel W. A. Noble developed the helper function with the help of Yefeng Yang and Shinichi Nakagawa. All authors read, commented on and edited the manuscript and approved the final submission.

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CONFLICT OF INTEREST STATEMENT

The authors reported no conflicts of interest.

PEER REVIEW

The peer review history for this article is available at https://www.webofscience.com/api/gateway/wos/peer-review/10.1111/2041-210X.14377.

DATA AVAILABILITY STATEMENT

All data (including analytical script) can be found at GitHub repository (https://github.com/Yefeng0920/WLS_RVE) and Zenodo (https://zenodo.org/records/11480297; Yang, 2024). A webpage showing the implementation of the proposed method in combination with a visualization tool can be accessed via https://yefeng0920.github.io/BiasRobustMA_tutorial/.

TRANSPARENCY AND OPENNESS

The authors affirm that the manuscript is an honest, accurate and transparent account of the study being reported, and no important aspects of the study have been omitted. Data, materials and code for replicating the analyses and figures are archived at our GitHub repository (https://github.com/Yefeng0920/WLS_RVE) and Zenodo (https://zenodo.org/records/11480297). A webpage showing the implementation of the proposed method in combination with a visualization tool can be accessed at https://yefeng0920.github.io/BiasRobustMA_tutorial/. The present research was not preregistered.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

Figure S1: Comparison of the mean population effect size estimates $\hat{\beta}$ and their standard errors $\mathrm{SE}(\hat{\beta})$ obtained from different meta-analytic models.

Figure S2: Paired comparison of the estimated mean population effect size estimates $\hat{\beta}$ of different effect size measures.

Figure S3: The relationship between the difference in the population mean effect size estimates $\hat{\beta}$ and the severity of publication bias.

Figure S4: Comparison of the population mean effect size estimates $\hat{\beta}$ and their standard errors $\mathrm{SE}(\hat{\beta})$ obtained from different meta-analytic models.

Figure S5: Paired comparison of the estimated overall effects $\hat{\beta}$ obtained from the benchmark method (MLMA), the first-step bias-robust model of the proposed approach (FE + VCV), and PET-PEESE adjustment method.

Table S1: Quantitative comparison of the population mean effect size estimates $\hat{\beta}$ obtained from different meta-analytic models.

Table S2: Quantitative relationship between the difference in the population mean effect size estimates $\hat{\beta}$ and the severity of publication bias.

Table S3: Quantitative comparison of the standard error $\mathrm{SE}(\hat{\beta})$ of population mean effect size estimates $\hat{\beta}$ obtained from different meta-analytic models.

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APPENDIX A

Meta-analytic model via linear mixed-effects model framework

Consider a meta-analysis encompassing a collection of J primary studies, where j-th study reports n_j effect size estimates y_{ij} and sampling variances s_{ij}^2 (where $i = 1, \dots, n_j$ and $j = 1, \dots, J$). Although our main focus is on the intercept-only meta-analysis model, let $\mathbf{x}_{ij}$, without loss of generality, be a row vector of p predictors (also known as, covariates or moderators) that induce systematic variations among the effect sizes. In the frequentist framework, $\mathbf{x}_{ij}$ is the so-called fixed effects (in our case, $\mathbf{x}_{ij}$ is an intercept term and thus equals 1). Likewise, let $\mathbf{z}_{ij}$ be a row vector of q predictors that lead to random variations among the effect sizes and are therefore considered random effects. Using the (generalized) linear mixed-model framework (Platt et al., 1999; Stram, 1996), the FE, and RE models and their more complex variants can be unified as a general form:

$$y_{ij} = \mathbf{x}_{ij}\boldsymbol{\beta} + \mathbf{z}_{ij}b_{ij} + e_{ij}, \quad \text{(A1)}$$

where $\boldsymbol{\beta}$ denotes the vector of regression coefficients for fixed-effects predictors $\mathbf{x}_{ij}$, representing the change in the (predicted) y_{ij} resulting from each one-unit change in $\mathbf{x}_{ij}$; b_{ij} denote the regression coefficients for random-effects predictors $\mathbf{z}_{ij}$, indicating (residual) variation in the effect sizes; e_{ij} denote the sampling error corresponding to y_{ij}.

For the sake of brevity, we stack effect size estimates y_{ij} for each j-th study and express Equation (1) in the matrix notation as Equation (A2), which is known as a specification of seemingly unrelated regressions (SURs) (Zellner, 1962):

$$\mathbf{y}_j = \mathbf{X}_j\boldsymbol{\beta} + \mathbf{Z}_j\boldsymbol{b}_j + \mathbf{e}_j, \quad \text{(A2)}$$

where $\mathbf{y}_j = \left(y_{1j}, \dots, y_{ij}, \dots, y_{n_j j}\right)'$ denotes a $n_j \times 1$ vector of effect size estimates, $\mathbf{X}_j$ is a $n_j \times p$ predictor matrix of p fixed effects with a $p \times 1$ vector of regression coefficient $\boldsymbol{\beta}$ (in our case, a vector composed solely of 1), $\mathbf{Z}_j$ is a $n_j \times q$ design matrix of q random effects with a $q \times 1$ vector of regression coefficient $\boldsymbol{b}_j = \left(b_{1j}, \dots, b_{ij}, \dots, b_{qj}\right)'$ and $\mathbf{e}_j = \left(e_{1j}, \dots, e_{ij}, \dots, e_{n_j j}\right)'$ represents a $n_j \times 1$ vector of sampling error effects for j-th study. The typical assumptions of meta-analytic modelling indicate $\mathrm{E}\left(\boldsymbol{b}_j\right) = \mathbf{0}$ and $\mathrm{Var}\left(\boldsymbol{b}_j\right) = \mathbf{G}_j$, where $\mathbf{G}_j$ denotes a $q \times q$ symmetric positive-definite variance–covariance matrix with $\mathrm{Var}\left(b_{ij}\right) = \sigma_i^2$ as the diagonal elements ($i = 1, \dots, q$), and $\mathrm{Cov}\left(b_{ij}, b_{hj}\right) = \varphi_{ihj}\sigma_{ij}\sigma_{hj}$ as the off-diagonal elements (where φ_{ihj} is the correlation between random effects b_{ij} and b_{hj}). Likewise, $\mathrm{E}\left(\boldsymbol{e}_j\right) = \mathbf{0}$ and $\mathrm{Var}\left(\boldsymbol{e}_j\right) = \mathbf{S}_j$ where $\mathbf{S}_j$ denotes a $n_j \times n_j$ symmetric positive-definite variance–covariance matrix with $\mathrm{Var}\left(e_{ij}\right) = s_{ij}^2$ as the diagonal elements and $\mathrm{Cov}\left(e_{ij}, e_{hj}\right) = \rho_{ihj}s_{ij}s_{hj}$ as the off-diagonal elements (where ρ_{ihj} is the sampling correlation or within-study correlation between two paired effect size estimates y_{ij} and y_{hj}) (Searle, 1968). Under the frequentist framework, Equation (A3) can be expressed as a marginal form of

$$\mathbf{y}_j \sim \mathrm{MVN}\left(\mathbf{X}_j\boldsymbol{\beta}, \boldsymbol{\Sigma}_j\right), \quad \text{(A3)}$$

where $\boldsymbol{\Sigma}_j = \mathrm{Var}\left(\mathbf{Z}_j\boldsymbol{b}_j + \mathbf{e}_j\right) = \mathbf{Z}_j\mathbf{G}_j\mathbf{Z}_j{}' + \mathbf{S}_j$ denotes a marginal or model-implied variance–covariance matrix of the effect size estimate $\mathbf{y}_j$. The specification of $\boldsymbol{\Sigma}_j$ explicitly reflects the true dependence structure of a meta-analytic dataset.

Minimum-variance unbiased estimator

The main 'estimands' of interest in a meta-analysis are the fixed-effects regression coefficient $\boldsymbol{\beta}$ and their uncertainties, including sampling error $\mathrm{SE}(\boldsymbol{\beta})$ (square root of sampling variance–covariance $\mathrm{Var}(\boldsymbol{\beta})$) or confidence intervals. The fixed-effects regression coefficient $\boldsymbol{\beta}$ with the minimum mean squared error can be obtained by using iterative generalized lease squares (GLS) or equivalently restricted maximum likelihood estimator (REML) (Van Houwelingen et al., 2002):

$$\hat{\boldsymbol{\beta}} = \left(\mathbf{X}'\mathbf{W}\mathbf{X}\right)^{-1}\mathbf{X}'\mathbf{W}\mathbf{y} = \left(\sum_{j=1}^{J}\mathbf{X}_j'\mathbf{W}_j\mathbf{X}_j\right)^{-1}\sum_{j=1}^{J}\mathbf{X}_j'\mathbf{W}_j\mathbf{y}_j, \quad \text{(A4)}$$

where $\mathbf{W}$ is an $n_jJ \times n_jJ$ matrix of weights of a collection of J primary studies, $\mathbf{W}_j$ is an $n_j \times n_j$ matrix of weights of jth study.

The associated sampling variance–covariance matrix of $\hat{\boldsymbol{\beta}}$ can be estimated with

$$\begin{aligned}\mathrm{Var}\left(\hat{\boldsymbol{\beta}}\right) &= \left(\mathbf{X}'\mathbf{W}\mathbf{X}\right)^{-1}\mathbf{X}'\mathbf{W}\boldsymbol{\Sigma}\mathbf{X}\left(\mathbf{X}'\mathbf{W}\mathbf{X}\right)^{-1} \\ &= \left(\sum_{j=1}^{J}\mathbf{X}_j'\mathbf{W}_j\mathbf{X}_j\right)^{-1}\sum_{j=1}^{J}\mathbf{X}_j'\mathbf{W}_j\boldsymbol{\Sigma}_j\mathbf{W}_j\mathbf{X}_j\left(\sum_{j=1}^{J}\mathbf{X}_j'\mathbf{W}_j\mathbf{X}_j\right)^{-1},\end{aligned} \quad \text{(A5)}$$

Since $\mathrm{E}\left(\mathbf{y}_j \mid \mathbf{X}_j\right) = \mathbf{X}_j\boldsymbol{\beta}$, Equation (A4) provides an unbiased estimator of $\boldsymbol{\beta}$, regardless of the specifications of the $n_j \times n_j$ weighting matrix $\mathbf{W}_j$:

$$\mathrm{E}\left(\hat{\boldsymbol{\beta}}\right) = \left(\sum_{j=1}^{J}\mathbf{X}_j'\mathbf{W}_j\mathbf{X}_j\right)^{-1}\sum_{j=1}^{J}\mathbf{X}_j'\mathbf{W}_j\mathrm{E}\left(\mathbf{y}_j\right) = \left(\sum_{j=1}^{J}\mathbf{X}_j'\mathbf{W}_j\mathbf{X}_j\right)^{-1}\sum_{j=1}^{J}\mathbf{X}_j'\mathbf{W}_j\mathbf{X}_j\boldsymbol{\beta} = \boldsymbol{\beta}, \quad \text{(A6)}$$

To obtain a minimum-variance unbiased estimator of $\boldsymbol{\beta}$, we need to pick a weighting matrix that produces the minimum $\mathrm{Var}\left(\hat{\boldsymbol{\beta}}\right)$ among all unbiased estimators (in this case, best linear unbiased prediction BLUP). Based on the generalized Gauss-Markov theorem, setting $\mathbf{W}_j = \boldsymbol{\Sigma}^{-1}$ leads to the unique solution to minimize $\mathrm{Var}\left(\hat{\boldsymbol{\beta}}\right)$:

$$\begin{aligned}\mathrm{Var}\left(\hat{\boldsymbol{\beta}}\right) &= \left(\sum_{j=1}^{J}\mathbf{X}_j'\mathbf{W}_j\mathbf{X}_j\right)^{-1}\sum_{j=1}^{J}\mathbf{X}_j'\mathbf{W}_j\boldsymbol{\Sigma}_j\boldsymbol{\Sigma}_j^{-1}\mathbf{X}_j\left(\sum_{j=1}^{J}\mathbf{X}_j'\mathbf{W}_j\mathbf{X}_j\right)^{-1} \# \\ &= \left(\sum_{j=1}^{J}\mathbf{X}_j'\mathbf{W}_j\mathbf{X}_j\right)^{-1} = \left(\sum_{j=1}^{J}\mathbf{X}_j'\boldsymbol{\Sigma}_j^{-1}\mathbf{X}_j\right)^{-1},\end{aligned} \quad \text{(A7)}$$

Setting $\mathbf{W}_j = \boldsymbol{\Sigma}_j^{-1}$ is the default inverse marginal variance–covariance weighting scheme in the meta-analytic models. Using $\mathbf{W}_j = \boldsymbol{\Sigma}_j^{-1}$ as weights require the knowledge of the dependence structure $\boldsymbol{\Sigma}_j = \mathbf{Z}_j\mathbf{G}_j\mathbf{Z}_j{}' + \mathbf{S}_j$. Methodological innovations offer various variance–covariance structures that can be imposed on $\mathbf{G}_j$ and $\mathbf{S}_j$ to capture the underlying dependence structure, consequently providing an efficient $\mathbf{W}_j$. As such, Equations (A4) and (A7) provide efficient estimation of $\boldsymbol{\beta}$ and unbiased $\mathrm{Var}\left(\hat{\boldsymbol{\beta}}\right)$ (e.g. nominal Type I error rates), respectively. However, both Equations (A4) and (A7) become invalid when the meta-analytic dataset is non-representative, for instance, due to selective reporting and statistically dependent effect sizes.

Unrestricted weighted least squares (UWLS) model

The UWLS model is a weighted linear regression used to address heteroskedasticity (i.e. an unequal variance of observation) in ordinary least squares regression. The UWLS was initially conceptualized as a statistical model for meta-analysis in particle physics (Baker & Jackson, 2013; Rosenfeld, 1975). Stanley and his colleagues subsequently fleshed out its theories in meta-analysis and termed it as unrestricted weighted least squares (Stanley et al., 2022; Stanley & Doucouliagos, 2015; Stanley & Doucouliagos, 2017). Unlike the FE model, the UWLS model relaxes the assumption that sampling variance s_{ij} is precisely known without uncertainty, instead assuming that s_{ij} is only known up to a proportionality constant σ_e^2. This results in a weighting scheme of $\mathbf{W}_j = \boldsymbol{\Sigma}_j^{-1} = \sigma_e^2 \mathbf{S}_j^{-1}$ in the UWLS model. The estimator of $\hat{\beta}$ is identical to Equation (10) in FE model, while the estimator of $\mathrm{Var}\left(\hat{\beta}\right)$ becomes

$$\mathrm{Var}\left(\hat{\beta}_{\mathrm{UWLS}}\right) = \frac{1}{\sigma_e^2 \sum_{j=1}^{J} \sum_{i=1}^{n_j} \left(1/s_{ij}\right)}, \tag{A8}$$

The term σ_e^2 technically refers to the residual variance or weighted mean squared error. This parameter can be estimated from data in contrast to the fixed value of 1 in FE and RE models, which is why the UWLS model is called the 'unrestricted' WLS (Stanley & Doucouliagos, 2015). The term σ_e^2 is beneficial to statistical inference on $\hat{\beta}_{\mathrm{UWLS}}$ at two aspects. On the one hand, it represents the overdispersion of effect sizes and thus accounts for heterogeneity in a multiplicative way (Stanley & Doucouliagos, 2015). This is why UWLS is also known as the 'multiplicative' method for meta-analysis. On the other hand, σ_e^2 can act as a scaling factor of $\mathrm{Var}\left(\hat{\beta}_{\mathrm{UWLS}}\right)$ to account for the uncertainty in estimating $\boldsymbol{\Sigma}_j^{-1}$ and improve statistical inference (Van Aert & Jackson, 2019). Interestingly, the estimator of $\mathrm{Var}\left(\hat{\beta}_{\mathrm{UWLS}}\right)$ in UWLS model with $1/\left(\tau^2 + s_{ij}^2\right)$ as weight is mathematically equivalent to Hartung–Knapp–Sidik–Jonkman adjustment to the random-effects meta-analysis (Hartung, 1999; Knapp & Hartung, 2003; Sidik & Jonkman, 2002).